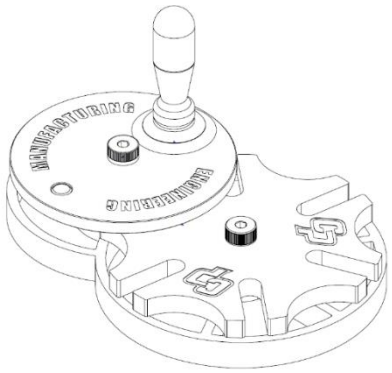
IME 244 Geneva Wheel Project



Overview:

The Cal Poly Geneva Wheel Project has been designed to give manufacturing engineering students experience modeling and manufacturing individual parts that fit together in an assembly using both traditional and non-traditional manufacturing processes. This lab project will occupy the first five and a half weeks of the IME 244 labs.

The most basic learning objectives are to:

1. Model and analyze a complex mechanical assembly using a CAD system.
2. Manufacture parts which assemble together into a mechanical mechanism.
3. Understand fits and describe how they are utilized in design and manufacturing.
4. Interpret Geometric Dimensioning & Tolerancing (GD&T) callouts on machined parts.
5. Determine the amount of "tolerance stack-up" on an assembly of parts.
6. Apply all the manufacturing skills, techniques, and theory discussed to manufacture an assembly of mechanical parts that fit together and function.
7. Describe the advantages and limitations of non-traditional machining processes.

The class will work on designing, manufacturing, and inspecting one part of the assembly per week for the next five weeks at the end of each IME 244 lab. Each week we will have an in class assignment pertaining to manufacturing the Geneva Wheel Assembly!

Lab Week #1 – Geneva Drive Wheel Mechanical Assembly Fits & CAD Models

Lab Week #2 – Geneva Drive Wheel CNC Machining (2- & 3-axis) & CAM Programming

Lab Week #3 – Geneva Drive Handle Injection Molding, Mold Design, Molding, & Machining

Lab Week #4 – Geneva Wheel Laser Engraving & Cutting

Lab Week #5 – Geneva Wheel Base Waterjet Cutting

Lab Week #5½ Geneva Wheel Assembly & Tolerance Stack Up Calculations